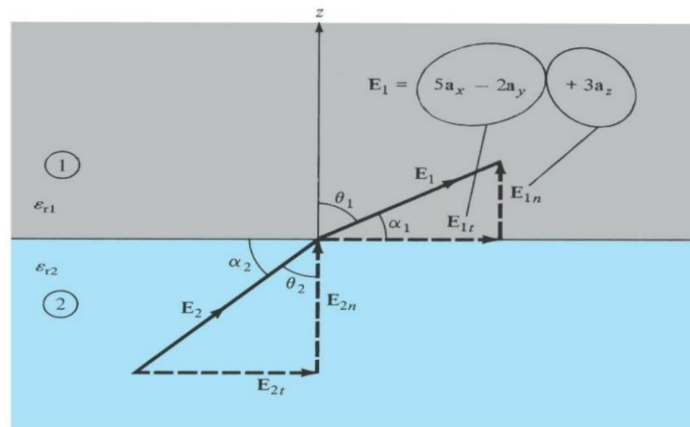




Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Worksheet 6-Solutions

1. Let the problem be as illustrated in the following figure:



a. Since $\mathbf{a}_z$ is normal to the boundary plane, we obtain the normal components as:

$$E_{1n} = \mathbf{E}_1 \cdot \mathbf{a}_n = \mathbf{E}_1 \cdot \mathbf{a}_z = 3$$

$$\mathbf{E}_{1n} = 3\mathbf{a}_z$$

$$E_{2n} = (\mathbf{E}_2 \cdot \mathbf{a}_z) \mathbf{a}_z$$

also:

$$\mathbf{E} = \mathbf{E}_n + \mathbf{E}_t$$

hence:

$$\mathbf{E}_{1t} = \mathbf{E}_1 - \mathbf{E}_{1n} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

thus:

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

Similarly:

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \rightarrow \epsilon_{r2}\mathbf{E}_{2n} = \epsilon_{r1}\mathbf{E}_{1n}$$

or:

$$\mathbf{E}_{2n} = \frac{\epsilon_{r1}}{\epsilon_{r2}}\mathbf{E}_{1n} = \frac{4}{3}(3\mathbf{a}_z) = 4\mathbf{a}_z$$

thus:

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}$$

$$\mathbf{E}_2 = 5\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z \text{ kV/m}$$

b. Let α_1 and α_2 be the angle of $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface while θ_1 and θ_2 are the angles they make with the normal to the interface as shown in the previous figure; that is:

$$\alpha_1 = 90 - \theta_1$$

$$\alpha_2 = 90 - \theta_2$$

since:

$$E_{1n} = 3$$

and

$$E_{1t} = \sqrt{25 + 4} = \sqrt{29}$$

$$\tan \theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3} = 1.795 \rightarrow \theta_1 = 60.9^\circ$$

hence:

$$\alpha_1 = 29.1^\circ$$

Alternatively:

$$\mathbf{E}_1 \cdot \mathbf{a}_n = |\mathbf{E}_1| \cdot 1 \cdot \cos \theta_1$$

or:

$$\cos \theta_1 = \frac{3}{\sqrt{38}} = 0.4867 \rightarrow \theta_1 = 60.9^\circ$$

Similarly:

$$E_{2n} = 4 \text{ and } E_{2t} = E_{1t} = \sqrt{29}$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{29}}{4} = 1.346 \rightarrow \theta_2 = 53.4^\circ$$

hence:

$$\alpha_2 = 36.6^\circ$$

Note that:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}$$

is satisfied.

c. The energy densities are given by:

$$w_{E1} = \frac{1}{2} \varepsilon_1 |\mathbf{E}_1|^2 = \frac{1}{2} \cdot 4 \cdot \frac{10^{-9}}{36\pi} \cdot (25 + 4 + 9) \cdot 10^6 = 672 \mu J/m^3$$

$$w_{E2} = \frac{1}{2} \varepsilon_2 |\mathbf{E}_2|^2 = \frac{1}{2} \cdot 3 \cdot \frac{10^{-9}}{36\pi} \cdot (25 + 4 + 16) \cdot 10^6 = 597 \mu J/m^3$$

d. At the center (3,4,-5) of the cube of side 2 m, $z = -5 < 0$; that is, the cube is in region 2 with $2 \leq x \leq 4$, $3 \leq y \leq 5$, $-6 \leq z \leq -4$. Hence:

$$W_E = \int w_{E2} dv = \int_{x=2}^4 \int_{y=3}^5 \int_{z=-6}^{-4} w_{E2} dx dy dz = w_{E2}(2)(2)(2) = 597 * 8 \mu J = 4.776 mJ$$

2. Assuming that conductors 1 and 2, respectively, carry $+Q$ and $-Q$ uniformly distributed on them. By applying Gauss's law to an arbitrary Gaussian cylindrical surface of radius ρ ($a < \rho < b$). Thus:

$$Q = \varepsilon \int \mathbf{E} \cdot d\mathbf{S} = \varepsilon E_{\rho} 2\pi\rho L$$

hence:

$$\mathbf{E} = \frac{Q}{2\pi\varepsilon\rho L} \mathbf{a}_{\rho}$$

Neglecting flux fringing at the cylinder ends:

$$V = - \int_a^b \mathbf{E} d\mathbf{l} = - \int_a^b \left[\frac{Q}{2\pi\varepsilon\rho L} \mathbf{a}_{\rho} \right] \cdot d\rho \mathbf{a}_{\rho}$$

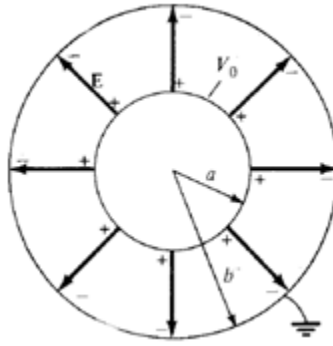
$$V = \frac{Q}{2\pi\varepsilon L} \ln \frac{b}{a}$$

Thus, the capacitance of a coaxial cylinder is given by:

$$C = \frac{Q}{V} = \frac{2\pi\varepsilon L}{\ln \frac{b}{a}}$$

3. Consider the spherical shells shown in the following figure. V depends only r and hence Laplace's equation becomes:

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dV}{dr} \right] = 0$$



Since $r \neq 0$ in the region of interest, we multiply through r^2 to obtain:

$$\frac{d}{dr} \left[r^2 \frac{dV}{dr} \right] = 0$$

integrating once gives:

$$r^2 \frac{dV}{dr} = A$$

or:

$$\frac{dV}{dr} = \frac{A}{r^2}$$

Integrating again gives:

$$V = -\frac{A}{r} + B$$

As usual, constants A and B are determined from the boundary conditions.

When:

$$r = b$$

$$V = 0 \rightarrow 0 = -\frac{A}{r} + B$$

or:

$$B = \frac{A}{b}$$

hence:

$$V = A \left[\frac{1}{b} - \frac{1}{r} \right]$$

also, when

$$r = a$$

$$V = V_o \rightarrow V_o = A \left[\frac{1}{b} - \frac{1}{a} \right]$$

or:

$$A = \frac{V_o}{\left[\frac{1}{b} - \frac{1}{a} \right]}$$

thus:

$$V = V_o \frac{\left[\frac{1}{r} - \frac{1}{b} \right]}{\frac{1}{a} - \frac{1}{b}}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{dr} \mathbf{a}_r$$

$$\mathbf{E} = -\frac{A}{r^2} \mathbf{a}_r = \frac{V_o}{r^2 \left[\frac{1}{a} - \frac{1}{b} \right]} \mathbf{a}_r$$

$$Q = \int \varepsilon \mathbf{E} \cdot d\mathbf{S} = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{\varepsilon_o \varepsilon_r V_o}{r^2 \left[\frac{1}{a} - \frac{1}{b} \right]} r^2 \sin \theta \, d\phi d\theta$$

$$Q = \frac{4\pi\epsilon_o\epsilon_r V_o}{\frac{1}{a} - \frac{1}{b}}$$

The capacitance is easily determined as:

$$C = \frac{Q}{V_o} = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$$

Substituting $a = 0.1 \text{ m}$, $b = 0.3 \text{ m}$, $V_o = 100 \text{ V}$ yields:

$$V = 100 \frac{\left[\frac{1}{r} - \frac{10}{3}\right]}{10 - \frac{10}{3}} = 15 \left[\frac{1}{r} - \frac{10}{3}\right] \text{ V}$$

Check:

$$\nabla^2 V = 0$$

$$V(r = 0.3\text{m}) = 0$$

$$V(r = 0.1\text{m}) = 100$$

$$\mathbf{E} = \frac{100}{r^2 \left[10 - \frac{10}{3}\right]} \mathbf{a}_r = \frac{15}{r^2} \mathbf{a}_r \text{ V/m}$$

$$Q = \pm 4\pi \cdot \frac{10^{-9}}{36\pi} \cdot \frac{(2.5) \cdot (100)}{10 - \frac{10}{3}} = \pm 4.167 \text{ nC}$$

The positive charge is induced on the inner shell; the negative charge is induced on the outer shell. Also:

$$C = \frac{|Q|}{V_o} = \frac{4.167 \times 10^{-9}}{100} = 41.67 \text{ pF}$$

